

Deep Research: Avatrada 57 Topic 024

Engineering Report: Volatility Targeting and Scaling

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Subject: Deep-Dive on Volatility Scaling for Position Sizing (ATR/IV)

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1.0 Executive Summary

This report provides a rigorous engineering analysis of the "Volatility Targeting" position sizing methodology as specified in the source document. The core objective of this technique is to normalize position risk across different assets, ensuring each contributes a consistent, predefined amount of risk to the portfolio. We will deconstruct the mechanisms for both equities (using Average True Range - ATR) and options (using Implied Volatility - IV), provide a detailed implementation strategy, and conduct a critical analysis of the system's potential failure modes, limitations, and areas for optimization. The central challenge addressed is the normalization of an option's Implied Volatility (a percentage) into a concrete dollar-risk equivalent for accurate sizing.

2.0 Technical Deconstruction

Volatility targeting is a risk management framework that sizes positions inversely to their recent or expected volatility. The fundamental principle is: **higher volatility leads to a smaller position size, and lower volatility leads to a larger position size**, for the same target risk budget.

2.1 Equity Position Sizing (ATR-Based)

For equities, the system uses Average True Range (ATR), a measure of *realized* (historical) price volatility.

2.1.1 Core Components & Formula

The provided formula is: $\text{Shares} = (\text{Account_Value} * \text{Target_Risk_Pct}) / (\text{ATR} * \text{Multiplier})$

Let's break down each component:

- **Account_Value** : Total market value of the trading account (e.g., \$1,000,000).
- **Target_Risk_Pct** : The percentage of the account value to be risked on a single position. For example, 1% (0.01).
- **Risk_Budget** : This is the numerator ($\text{Account_Value} * \text{Target_Risk_Pct}$). It represents the maximum acceptable loss in dollar terms for this position if the asset moves against the trade by a "standard" volatile amount (e.g., \$10,000).
- **ATR (Average True Range)**: The primary volatility metric. It is the moving average (typically 14-day) of the True Range.
 - **True Range (TR)** is the greatest of the following:
 1. $\text{Current High} - \text{Current Low}$
 2. $\text{abs}(\text{Current High} - \text{Previous Close})$
 3. $\text{abs}(\text{Current Low} - \text{Previous Close})$
 - ATR is expressed in dollars (e.g., \$2.50 for a stock), representing the average daily price range, accounting for overnight gaps.
- **Multiplier** : A scalar used to adjust the risk denominator. A multiplier of 1.0 means you are sizing the position based on a 1-ATR move. A multiplier of 2.0 means you are sizing for a 2-ATR move, effectively halving the position size and the risk taken. This is a crucial parameter for tuning the system's overall risk appetite.

2.1.2 Mathematical Logic

The formula can be re-written to highlight its logic:

```
# Total dollar risk allowed for the position
Risk_Budget_USD = Account_Value * Target_Risk_Pct

# The expected dollar move per share that defines our risk event
Risk_Per_Share_USD = ATR * Multiplier

# How many shares can we hold so that a 'Risk_Per_Share' event equals our
'Risk_Budget'?
Number_of_Shares = Risk_Budget_USD / Risk_Per_Share_USD
```

2.2 Option Position Sizing (IV-Based)

Options are derivative instruments, and their risk profile is multi-faceted (the "Greeks"). The challenge is converting Implied Volatility (IV), an annualized percentage, into a practical dollar-risk metric for a single options contract.

2.2.1 Normalizing Implied Volatility to Dollar Risk

IV represents the market's expectation of the annualized 1-standard-deviation move of the *underlying asset*. A direct comparison to ATR is not possible. We must translate this percentage into the expected dollar price change of the *option contract itself* over a short time horizon (e.g., one day).

A robust method involves using the option's Greeks, primarily **Delta** and **Gamma**, to approximate this change.

1. **Calculate the Expected 1-Day Move of the Underlying:** First, we de-annualize the IV to find the expected 1-standard-deviation move for a single trading day.

```latex

**S = Underlying Price, IV = Implied Volatility (e.g., 0.30 for 30%)**

---

**T\_days = Number of trading days in a year (approx. 252)**

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$\text{Expected\_1\_Day\_Underlying\_Move\_USD} = S * (IV / \sqrt{T\_days})$  `` For a \$100 stock with 30% IV, this is  $\$100 * (0.30 / \sqrt{252}) \approx \$1.89$ . This means the market expects a daily move of about \$1.89 with ~68% probability.

2. **Estimate the Option Price Change (The Dollar-Risk Equivalent):** The option's price will not move by \$1.89. Its change is a function of its Greeks. We can use a second-order Taylor series expansion to estimate the option's price change for a given move in the underlying. This is the **Delta-Gamma Approximation**.

````latex

$\Delta P_{\text{option}} = \text{Change in Option Price}$

$\Delta S = \text{Change in Underlying Price}$ (from step 1)

Note: Gamma is per \$1 move, so the formula is $0.5 * \text{Gamma} * (\Delta S)^2$

$\Delta P_{\text{option}} \approx (\text{Delta} * \Delta S) + (0.5 * \text{Gamma} * (\Delta S)^2)$ `` This ΔP_{option} is our **Dollar-Risk Equivalent per Contract**. It represents the projected profit or loss on the option for a 1-standard-deviation adverse move in the underlying.

2.2.2 Final Option Sizing Formula

With the dollar risk per contract established, the sizing formula becomes analogous to the equity formula:

```
# Dollar risk per contract, as calculated above
Dollar_Risk_Per_Contract = abs( $\Delta P_{\text{option}}$ ) * 100_shares_per_contract

# Risk budget remains the same
Risk_Budget_USD = Account_Value * Target_Risk_Pct

# Final formula for number of contracts
Number_of_Contracts = Risk_Budget_USD / Dollar_Risk_Per_Contract
```

3.0 Implementation Strategy

This system will be implemented in Python, leveraging standard financial and data analysis libraries.

3.1 Required Libraries

```
pip install pandas numpy yfinance py_vollib
```

- **pandas**: For data manipulation and time-series analysis.
- **numpy**: For numerical operations.
- **yfinance**: For fetching stock and option data (for prototyping; a production system requires a premium data provider like Polygon.io or Bloomberg).
- **py_vollib**: A library for calculating option prices and Greeks.

3.2 Equity Sizing Implementation

```
import pandas as pd
import numpy as np
import yfinance as yf

def calculate_atr(data: pd.DataFrame, period: int = 14) -> float:
    """Calculates the Average True Range (ATR) for a given period."""
    high_low = data['High'] - data['Low']
    high_close = np.abs(data['High'] - data['Close'].shift())
    low_close = np.abs(data['Low'] - data['Close'].shift())

    tr = pd.concat([high_low, high_close, low_close], axis=1).max(axis=1)
    atr = tr.ewm(alpha=1/period, adjust=False).mean() # Using EMA for ATR

    return atr.iloc[-1]

def get_equity_position_size(
    symbol: str,
    account_value: float,
    target_risk_pct: float,
```

```

    atr_period: int = 14,
    atr_multiplier: float = 2.0
) -> float:
    """Calculates position size for an equity based on ATR volatility."""

    # 1. Define Risk Budget
    risk_budget_usd = account_value * target_risk_pct

    # 2. Fetch Data and Calculate ATR
    # Fetch enough data for ATR calculation (e.g., period + buffer)
    hist_data = yf.Ticker(symbol).history(period=f"{atr_period * 2}d")
    if hist_data.empty:
        raise ValueError(f"No data found for symbol {symbol}")

    current_atr = calculate_atr(hist_data, period=atr_period)

    # 3. Calculate Risk Per Share
    risk_per_share_usd = current_atr * atr_multiplier

    if risk_per_share_usd == 0:
        return 0 # Avoid division by zero

    # 4. Calculate Position Size
    num_shares = risk_budget_usd / risk_per_share_usd

    return np.floor(num_shares) # Return whole shares

# --- Example Usage ---
account_val = 1000000
risk_pct = 0.01 # 1% risk
shares_to_buy = get_equity_position_size("AAPL", account_val, risk_pct)
print(f"Calculated Position Size for AAPL: {shares_to_buy} shares")

```

3.3 Option Sizing Implementation

```

from py_vollib.black_scholes.greeks import analytical as greeks

def get_option_position_size(
    underlying_symbol: str,

```

```

    option_data: dict, # Expects {'K', 'T', 'r', 'sigma', 'flag'}
    account_value: float,
    target_risk_pct: float,
    trading_days_per_year: int = 252
) -> float:
    """
    Calculates position size for an option based on IV and Greeks.

    option_data contains:
    K: Strike Price
    T: Time to expiration in years
    r: Risk-free rate
    sigma: Implied Volatility (IV)
    flag: 'c' for call, 'p' for put
    """

    # 1. Define Risk Budget
    risk_budget_usd = account_value * target_risk_pct

    # 2. Get Underlying Price
    underlying = yf.Ticker(underlying_symbol)
    S = underlying.history(period="1d")['Close'].iloc[-1]

    # 3. Calculate Expected 1-Day Underlying Move
    iv = option_data['sigma']
    expected_move_usd = S * (iv / np.sqrt(trading_days_per_year))

    # 4. Calculate Greeks
    delta = greeks.delta(option_data['flag'], S, option_data['K'],
option_data['T'], option_data['r'], iv)
    gamma = greeks.gamma(option_data['flag'], S, option_data['K'],
option_data['T'], option_data['r'], iv)

    # 5. Calculate Dollar Risk per Contract (Delta-Gamma Approximation)
    # We assume an adverse move, so for a long call, the move is down. For a
long put, up.
    # For simplicity, we can use the absolute value of the price change.
    # The direction of the move should align with the trade direction for
accurate risk.
    # E.g., for a long call, risk is a downward move (-expected_move_usd).

```



```

    adverse_move = -expected_move_usd if option_data['flag'] == 'c' else
expected_move_usd

    option_price_change = (delta * adverse_move) + (0.5 * gamma *
(adverse_move**2))

    # Dollar risk is per share; multiply by 100 for a standard contract
    dollar_risk_per_contract = abs(option_price_change * 100)

    if dollar_risk_per_contract == 0:
        return 0

    # 6. Calculate Position Size
    num_contracts = risk_budget_usd / dollar_risk_per_contract

    return np.floor(num_contracts)

# --- Example Usage ---
# Fictional data for an AAPL call option
aapl_call_data = {
    'K': 175,          # Strike Price
    'T': 30/365.25,    # Time to expiration (30 days)
    'r': 0.05,         # Risk-free rate (5%)
    'sigma': 0.25,     # Implied Volatility (25%)
    'flag': 'c'        # It's a call
}

contracts_to_buy = get_option_position_size("AAPL", aapl_call_data,
account_val, risk_pct)
print(f"Calculated Position Size for AAPL Call: {contracts_to_buy} contracts")

```

4.0 Critical Analysis

While powerful, the Volatility Targeting model is not infallible. It is a model with specific assumptions and is subject to several failure modes and edge cases.

4.1 Potential Failure Modes & Edge Cases

1. **Non-Stationary Volatility:** The model's primary weakness. Both ATR (backward-looking) and IV (forward-looking but model-dependent) assume that the near-future volatility will resemble the measured volatility. A sudden event (e.g., earnings announcement, geopolitical news) can cause a regime shift, rendering the historical or implied metric obsolete. A position sized for low volatility can incur massive losses if volatility spikes unexpectedly.
2. **Gap Risk:** ATR incorporates gaps in its calculation, but an extreme "black swan" gap (e.g., a 5-ATR move overnight) will violate the model's risk budget. The `Multiplier` parameter provides a buffer, but it cannot protect against tail events.
3. **Liquidity Constraints & Market Impact:** The model is blind to market liquidity. It may calculate a position size of 50,000 shares for a stock that only trades 100,000 shares per day. Attempting to execute this order would cause significant slippage and market impact, altering the entry price and invalidating the risk calculation.
4. **Options-Specific Failures:**
 - **Gamma Risk near Expiration:** As an option approaches expiration, its Gamma can become extremely large. The Delta-Gamma approximation becomes unstable, and small moves in the underlying can cause massive, unpredictable swings in the option's value (and risk). The model may severely underestimate risk in the final days of an option's life.
 - **Vega Risk (Volatility Crush):** The model primarily sizes based on the underlying's price risk (Delta/Gamma). It does not explicitly budget for Vega risk—the risk of a collapse in Implied Volatility. After a major event like earnings, IV often plummets, which can cause large losses on long option positions even if the underlying price moves favorably.

- **Data Integrity:** Option chains can have erroneous IV or Greek values, especially for illiquid strikes. Sizing based on bad data will lead to incorrect risk allocation.

4.2 Optimizations and Enhancements

1. **Hybrid Volatility Models:** For equities, blend historical and implied volatility. For example, use the maximum of (14-day ATR, 30-day Implied Volatility converted to a daily dollar value). This makes the model more responsive to market expectations.
2. **Portfolio-Level Correlation Analysis: This is the most critical enhancement.** The described model sizes each position in isolation. It completely ignores the correlation between assets. Holding two highly correlated positions, each sized for 1% risk, does *not* result in a 2% portfolio risk; the combined risk is likely closer to 2% on each, for a total of nearly 4% during an adverse market move. A robust system must use a covariance matrix to calculate the marginal contribution to risk of a new position and size it accordingly to maintain a target *portfolio* volatility.
3. **Dynamic Multiplier:** The `Multiplier` can be dynamic. In high-uncertainty environments (e.g., $VIX > 30$), the multiplier could be increased (e.g., from 2.0 to 3.0) to systematically reduce risk across the board.
4. **Liquidity-Aware Sizing:** The final position size should be capped as a percentage of the average daily volume (ADV). For example, `final_size = min(calculated_shares, 0.02 * ADV)`. This prevents the model from establishing untenable positions.
5. **Stress Testing:** The model's risk assumptions should be regularly stress-tested against historical crises (e.g., 2008, 2020) to understand how it would behave under extreme market conditions. This helps in refining the `Multiplier` and other risk control parameters.